

[N.B. – Answer all the questions. Each question carries ONE mark. Block fully, with a black ball- point pen, the circle of the letter that stands for the correct/best answer in the “Answer sheet” for the Multiple Choice Questions Examination.]

Candidates are asked not to leave any mark or spot on the question paper.

1. A survey categorizing people by their favorite color is an example of which measurement scale?
- (a) Nominal

(b) Ordinal

(c) Interval

(d) Ratio
2. Which is a discrete variable?
- (a) Number of languages spoken by a person

(b) Time taken to complete a race

(c) Length of a road

(d) Volume of water in a tank
3. If $x_1 = 3$, $x_2 = 2$, $x_3 = -6$, and $x_4 = 4$, what is $\sum_{i=1}^4 x_i^2$?
- (a) 45

(b) 65

(c) 85

(d) 89

Answer the next TWO questions based on the following information

The table below shows monthly sales (in thousands of dollars) for two products in four regions.

Product	North	South	East	West
A	12	15	10	8
B	18	22	14	16

4. Calculate total sales of Product A across all regions.
- (a) 40

(b) 42

(c) 45

(d) 48
5. Calculate total sales of all products across all regions.
- (a) 95

(b) 100

(c) 105

(d) 115
6. If each observation is multiplied by 3 and then increased by 4, the arithmetic mean of the original data will be:
- (a) Multiplied by 3

(b) Increased by 4

(c) Multiplied by 3 and then increased by 4

(d) Unchanged
7. The following are values of arithmetic, geometric, and harmonic mean, respectively
- i. {10, 12, 14}

ii. {10, 10, 10}

iii. {10, 8, 6}
- Which one is possible?
- (a) i and ii

(b) i and iii

(c) ii and iii

(d) i, ii and iii
8. The values of mode and median of moderately skewed distribution are 60 and 65. What is value of the mean?
- (a) 70

(b) 67.5

(c) 72

(d) 55
9. Which of the following are true about the median?
- i. It is the middle value when data is ordered

ii. It is unaffected by extreme outliers

iii. It is always equal to the mean in a symmetrical distribution
- Which one is correct?
- (a) i and ii

(b) i and iii

(c) ii and iii

(d) i, ii and iii

10. Which of the following are true about standard deviation?

- i. It has the same unit as the original data
- ii. It is the square root of variance
- iii. A larger standard deviation indicates greater dispersion

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

11. If the standard deviation of a dataset is 4, what is its variance?

- (a) 2 (b) 4 (c) 8 (d) 16

12. If the mean of a dataset is 50 and standard deviation is 10, the coefficient of variation is:

- (a) 10% (b) 15% (c) 20% (d) 25%

13. If $Q_1 = 25$ and $Q_3 = 45$, the quartile deviation is:

- (a) 5 (b) 7.5 (c) 10 (d) 12.5

14. Which of the following does not depend on change of origin?

- i. Median
- ii. Standard deviation
- iii. Regression coefficient

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

Answer the next TWO questions based on the following information

For a frequency distribution, the following central moments are given:

$$\mu_2 = 4, \quad \mu_3 = -4, \quad \mu_4 = 48$$

15. What is the variance of the distribution?

- (a) 2 (b) 4 (c) 8 (d) 16

16. What is Karl Pearson's coefficient of skewness based on moments ($\sqrt{\beta_1}$)?

- (a) -1.0 (b) -0.5 (c) 0.5 (d) 1.0

17. For a mesokurtik distribution, $\beta_2 =$ --

- (a) 0 (b) -3 (c) 3 (d) 1

18. The correlation coefficient between two variables lies between:

- (a) 0 and 1 (b) -1 and 0 (c) -1 and +1 (d) $-\infty$ and $+\infty$

19. If the correlation coefficient between X and Y is 0.6, the correlation between 2X and 3Y is:

- (a) 0.6 (b) 0.2 (c) 1.8 (d) Cannot be determined

20. If the regression coefficients are 0.8 and 0.5, the correlation coefficient is:

- (a) 0.40 (b) 0.63 (c) 0.80 (d) 1.30